210 T0 Korrekturen Wicklung

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Contents

Preliminary Note

Status: Fully incorporated — May 2026

All clarifications listed in this document (W1–W6) have been incorporated into the respective source documents. A merge with Doc. 190 is no longer necessary, as both documents are retained as historical archives.
This document is **no longer to be used as a binding reference** — the current state of each source document is authoritative.

This document supplements the errata register Doc. 190 with clarifications that exclusively concern the **winding-number structure of the particles on the fractal torus**.

The statements recorded here have been incorporated into the affected documents. Older documents were *not* modified — the corrections were implemented in revised versions.

Core statement. Each fermion in FFGFT carries three *distinct* winding-number layers, which in the older documents appear under the same symbols (n, l, j) or "winding", but have different geometric meanings:

- a universal base winding number $f = 1/(4\xi) = 7500$ (the same for all particles),
- a spin winding (n_θ, n_ϕ) on the inner spin fibration of the 4D torus T^4,
- a generation winding as fractal branching with Hausdorff dimension p_{g-2}.

The space-time mode of each particle lives on the **4D torus** T^4 and carries four integer winding numbers (k_x, k_y, k_z, k_t). These four layers are clearly separated in what follows. A fifth refinement (W5) additionally clarifies the bridge formula between the direct geometric and the Yukawa mass formula and resolves the ambiguity of the symbol f_i in the older documents.

.1 Refinement W1: Universal base winding number $f = 7500$

Affected Documents

- Doc. 060 (Music parallels) §"Circle of fifths ↔ torus topology": "$f = 7500$ windings".
- Doc. 018 (Anomalous g-2) §"The real sub-Planck factor $f = 7500$".
- Doc. 149 (FFGFT torsion) §"The ideal anchor number": $f = 1/(4\xi) = 30000/4 = 7500$.

Binding Reading

$$\boxed{f = \frac{1}{4\xi} = \frac{30000}{4} = 7500} \tag{210.1}$$

f is the *winding density of the universal 4D torus* and is **not particle-specific**. All particles are modes on the same torus with the same base winding. Where older documents speak of "the winding number of a particle", this does not refer to f, but to one of the subordinate windings from W2 or W3.

Justification

The constant $f = 7500$ is the geometric inverse of 4ξ, hence directly derived from the geometry parameter $\xi_0 = 4/30000$ (Doc. 180, 182). Its prime factorization $7500 = 2^2 \cdot 3 \cdot 5^4$ (Doc. 149 §65) reflects the tetrahedral symmetry (3), the spatial dimensionality (2^2), and the pentagonal projection (5^4) – all universal structural features, not particle-specific ones.

.2 Torus Geometry: Symbol Reference

For the winding designations in W1–W4 to be unambiguous, here first the torus geometry and the symbol legend.

What "4D torus" means here

The FFGFT torus is a **four-dimensional torus** T^4. Important: time is not a separate, linear coordinate $\mathbb{R}$ *next to* the torus, but **the fourth compact circle direction** *within* the torus itself. This is the geometric expression of the time-mass duality $\tilde{T} \cdot m = 1$: time and the three spatial directions are circles of equal status, all four cyclically closed.

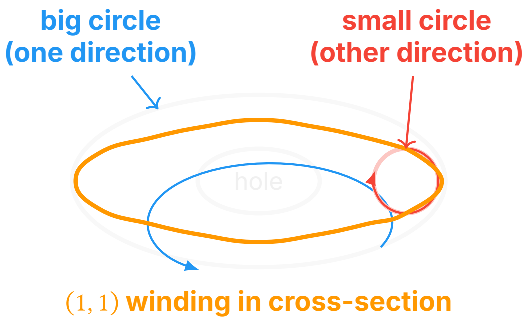
$$T^4 = S^1_x \times S^1_y \times S^1_z \times S^1_t \tag{210.0}$$

Hence *every* mode of the universe carries four integer winding numbers (k_x, k_y, k_z, k_t) – three spatial and one temporal.

Note on older formulations: where Doc. 202 §116 writes the support manifold as $T^3 \times \mathbb{R}$, the low-energy approximation is meant in which the temporal circle direction appears unrolled to $\mathbb{R}$. Binding is the full 4D torus structure T^4.

Anatomy of a Torus – Section View

Since T^4 is four-dimensional, it cannot be drawn directly. The following sketch shows a **2D cross-section** through two selected circle directions – e.g. (x, t) or (θ, ϕ). The other two circle directions are orthogonal and treated separately.



Reading aid for the sketch: the gray double-ellipse is the top view of a cross-section torus – the tube seen from above, with the hole in the middle. The red arrow on the right shows one circle traversed once (cross-section of the tube). The blue arrow shows the other circle traversed once (around the hole). The orange path is a $(1, 1)$ winding: a single simultaneous traversal of both circles.

On the 4D torus, every mode lives with *four* such winding numbers simultaneously. The sketch shows only two of them.

Symbol Legend

Symbol	Designation	Geometric meaning
T^4	4D space-time torus	Support manifold of FFGFT. $T^4 = S^1_x \times S^1_y \times S^1_z \times S^1_t$ – four cyclic circle directions, three spatial and one temporal. All four have the same topological status.
S^1_x, S^1_y, S^1_z	Spatial circle directions	The three spatial dimensions, each rolled into a circle of base period L_0.
S^1_t	Temporal circle direction	Time as the fourth circle direction. Period $\tilde{T}_0 = L_0/c$. *Not* a separate $\mathbb{R}$ line; time-mass duality $\tilde{T} \cdot m = 1$ is geometrically visible here.
(k_x, k_y, k_z, k_t)	Wave-vector winding numbers	Four integer winding numbers of a mode on T^4. Each counts how often the mode wraps around the corresponding circle direction.
θ, ϕ	Poloidal / toroidal angle	In the *cross-section* representation of the torus through two circle directions, the cross-section circle ("small circle") is denoted θ, the loop around the hole ("big circle") is denoted ϕ.
(n_θ, n_ϕ)	Spin winding	Pair of winding numbers specifically for the spin property of a particle. Encodes not the space-time mode, but the inner spin structure (W2).
L_0	Base period	Length scale $L_0 = \ell_P/\xi$ of the universal torus (Doc. 182). Defines the size of *all four* circle directions together with c.
f	Winding density	$f = 1/(4\xi) = 7500$. Number of base periods L_0 that fit into the Hubble distance; *not* a winding number of individual particles. The factor 4 in $1/(4\xi)$ comes directly from the four-dimensionality of T^4.

Space-time mode vs. spin winding

On the 4D torus T^4, every mode carries *two distinct sets* of winding numbers, independent of each other:

	Space-time mode	Spin winding
Lives on	All four circles of T^4	Inner spin fibration over T^4
Winding numbers	(k_x, k_y, k_z, k_t)	(n_θ, n_ϕ)
What they encode	Space-time motion of the mode, generation, particle identity (W4)	Spin (W2)
Characterizes	Generation and identity of the particle	Spin class: fermion / boson / Higgs

Both winding sets exist simultaneously. An electron sits with space-time wave vector (k_x, k_y, k_z, k_t) on T^4 *and* additionally carries the spin winding $(1, 1)$ on the inner fibration.

.3 Refinement W2: Spin winding (n_θ, n_ϕ)

Affected Documents

- Doc. 051 (Dirac, FFGFT) §"Topological interpretation", §"Winding numbers on the torus".
- Doc. 202 §"Quantum numbers of the modes" §7.4 (spin structure).

Binding Reading

The **spin winding** is a property on the inner spin fibration over the 4D torus T^4, with two independent winding numbers (n_θ, n_ϕ):

$$\text{Spin-}s \longleftrightarrow (n_\theta, n_\phi) \text{ with } \frac{n_\phi}{n_\theta} = 2s \tag{210.2}$$

Particle class	Spin s	(n_θ, n_ϕ)
All fermions (e, μ, τ, ν, q)	$1/2$	$(1, 1)$
Gauge bosons (γ, $W^\pm$, Z, g)	1	$(1, 2)$
Higgs boson	0	$(1, 0)$
Hypothetical graviton	2	$(1, 4)$

Geometric reading of the fermion's $(1, 1)$ winding: The orange path in the sketch above shows what it means to wind $(1, 1)$. Traversing the path goes *once around the small circle* ($n_\theta = 1$) and *once around the big circle* ($n_\phi = 1$). Only after *two* such traversals does a point with the same phase return to itself – this is the topological origin of the 720° periodicity of spin-$\frac{1}{2}$.

Important: the spin winding does *not* distinguish the twelve fermions from one another – they all carry $(1, 1)$. It only distinguishes spin classes.

Justification

The topological interpretation of spin as a winding number is explicitly developed in Doc. 051 and supported by the Clifford-algebra structure (Doc. 050, 051). It is independent of the generation hierarchy of masses and applies universally to all spin-$\frac{1}{2}$ fermions.

.4 Refinement W3: Generation winding as fractal branching

Affected Documents

- Doc. 018 (anomalous g-2 v12) §"Muon: fractal extra winding", §"Tau: more complex fractal structure".
- Doc. 149 (FFGFT torsion) Abstract.

Binding Reading

The generations differ by **fractal branchings** of the base winding with characteristic Hausdorff dimension $p_{g\text{-}2}$:

$$a_\ell = a_e^{\text{base}} + \frac{4\pi}{f^{\,p_\ell}} \tag{210.3}$$

Particle	$p_{g\text{-}2}$	Geometric meaning	Source
Electron	–	Base winding (tetrahedral projection)	Doc. 018 §6
Muon	$5/3$	Brownian motion in 2D ("partially branched winding")	Doc. 018 §7
Tau	$4/3$	Box-counting of the Koch curve (stronger branching)	Doc. 018 §8

Key property: higher generation ⇒ *lower* Hausdorff dimension ⇒ stronger fractal branching ⇒ larger contribution to the anomalous magnetic moment.

Justification

The Hausdorff dimensions $5/3$ and $4/3$ are explicitly introduced in Doc. 018 §215–290 and traced to known fractal structures (2D Brownian motion, Koch curve). Formula (210.3) reproduces a_e and a_μ with fractal correction $K_{\text{frak}}^{-3/2}$ to $0.014\,\%$ and $0.15\,\%$ accuracy, respectively (Doc. 018 §211, §268).

.5 Refinement W4: Three meanings of (n, l, j) – clean separation

Problem Statement

In the older documents the symbol triple (n, l, j) appears in *three* different meanings:

Source	Meaning	Values for electron, muon, tau
Doc. 005, 006, 042	Hydrogen-like generation indices	$(1, 0, \frac{1}{2})$, $(2, 1, \frac{1}{2})$, $(3, 2, \frac{1}{2})$
Doc. 005 §153 (fractal method)	Three independent indices (n_1, n_2, n_3), $n_{\text{eff}} = n_1 + n_2 + n_3$	$(1, 0, 0)$, $(2, 1, 0)$, $(3, 2, 0)$
Doc. 202 §72–88	Wave-vector components $n = k_x^2 + k_y^2 + k_z^2$, $l^2 = k_x^2 + k_y^2$	$\mathbf{k} = (1, 0, 0)$, $(1, 1, 0)$, $(1, 1, 1)$; hence $n = 1, 2, 3$

The first two tables assign values to the triple labels without specifying a geometric construction; Doc. 202 derives n *retrospectively* from the wave vector.

Binding Reading

Binding is the wave-vector definition on the 4D torus T^4. The spacetime mode carries a *four-dimensional* wave vector:

$$\boxed{\mathbf{k} = \frac{2\pi}{L_0}(k_x, k_y, k_z, k_t), \quad k_x, k_y, k_z, k_t \in \mathbb{Z}} \tag{210.4}$$

with n, l, j as derived quantities from the three *spatial* components:

$$n = k_x^2 + k_y^2 + k_z^2 \quad \text{(principal quantum number, spatial part)} \tag{210.5}$$

$$l = \text{orbital angular momentum from spherical harmonics } Y_{l,m}(\theta, \phi) \tag{210.6}$$

$$j = l \pm \tfrac{1}{2} \quad \text{(spin-orbit coupling, spin winding from W2)} \tag{210.7}$$

The temporal winding number k_t, after *Wick rotation* or low-energy unrolling, gives the mass energy of the mode:

$$E_t = \frac{2\pi k_t}{\tilde{T}_0} = \frac{2\pi k_t c}{L_0} \tag{210.7a}$$

In the low-energy limit $k_t \to 0$ the temporal circle direction unrolls onto $\mathbb{R}$ – this is the form in which Doc. 202 §116 writes the manifold as $T^3 \times \mathbb{R}$.

Note: the notation $l = \sqrt{k_x^2 + k_y^2}$ in Doc. 202 §85–90 is to be understood as a *scaling relation* and does not replace the exact quantization $l(l+1) = \langle k_x^2 + k_y^2 \rangle$. The integer values $(0, 1, 2)$ for l in the lepton table are the corresponding orbital angular momentum eigenvalues, not the result of direct square root extraction.

Language Convention for Future Documents

When meaning is	Designation to use
Wave-vector winding	$(k_x, k_y, k_z, k_t) \in \mathbb{Z}^4$
Hydrogen quantum numbers	(n, l, j)
Spin winding	(n_θ, n_ϕ)
Hausdorff branching exponent (g-2 anomaly)	$p_{g\text{-}2}$
Yukawa mass exponent	p_{Yuk} or simply p in the form $m = r\xi^p v$

Important: the symbols $p_{g\text{-}2}$ and p_{Yuk} denote *different* quantities with different values:

Particle	p_{Yuk} (mass formula)	$p_{g\text{-}2}$ (Hausdorff)
Electron	$3/2$	– (base)
Muon	1	$5/3$
Tau	$2/3$	$4/3$

.6 Refinement W5: Bridge formula between direct and Yukawa method

Affected Documents

- Doc. Teilchenmassen_De (older version), §"Complete quark analysis with both methods" and §"Corrected interpretation of mathematical equivalence".
- Doc. 005 (T0-tm-Extension), §"Method 1: Direct geometric resonance (lepton basis)": values $f_\mu = 207$, $f_\tau = 12.3$.

- Doc. 006 (T0-Particle Masses), §"Universal table": three different readings of the same symbol f_i.
- Doc. T0_Teilchenmassen_De (Z. 384), §"Mathematical proof of equivalence": identity claim $K_{\text{frak}}/(\xi_0 f_i) \cdot C_{\text{conv}} = r_i \, \xi_0^{p_i} \, v$ without explicit resolution of the bridge formula.

Problem Statement

In the older documents the symbol f_i is used in *three* different meanings, often not clearly separated:

Meaning	**Definition**	**Values for e, μ, τ**
$f_i^{\text{(cluster)}}$	Geometric cluster label in $\xi_i = \xi_0 \cdot f_i$	$1,\ 16/5,\ 25/9$ (rational fractions, identical with r_i)
$f_i^{\text{(bridge)}}$	Inverse frequency size $f_i = 1/(\xi_0 \cdot m_i)$	$1.47 \cdot 10^7$, $7.10 \cdot 10^4$, $4.22 \cdot 10^3$ (orders of magnitude 10^3–10^7)
$f_i^{\text{(ratio)}}$	Mass ratio m_i/m_e (in Doc. 005 erroneously entered as f)	$1,\ 207,\ 3477$ resp. $1,\ 207,\ 12.3$

The three meanings are *not* equivalent. The use of the same letter for three distinct quantities is a main source of confusion in the discussion of the "two methods".

Binding Reading

FFGFT uses as its canonical mass formula the Yukawa form (Doc. 006 §161):

$$\boxed{m_i = r_i \cdot \xi_0^{p_i} \cdot v} \tag{210.9}$$

with $r_i, p_i \in \mathbb{Q}$ from the consolidated table (above).
The direct geometric form

$$m_i = \frac{1}{\xi_0 \cdot f_i^{\text{(bridge)}}} \tag{210.10}$$

is linked to (210.9) by the **bridge formula** (Doc. Teilchenmassen_De Z. 649):

$$\boxed{f_i^{\text{(bridge)}} = \frac{1}{r_i \cdot \xi_0^{p_i+1} \cdot v}} \tag{210.11}$$

Hence the direct and Yukawa forms are **not two independent paths**, but *one and the same formula in two notations*, mutually convertible by the identity (210.11).

Character of the Equivalence

The older version (Doc. Teilchenmassen_De Z. 640) states this plainly:

> "The mathematical equivalence of both methods is given *by definition*, when the parameters (r_i or f_i) are determined from the same experimental masses. The equivalence is not a proof for the theory, but a consistency property of the mathematical structure."

Binding: statements like "the two methods confirm each other to 0.2% accuracy" (Doc. 032 §70) are *tautological* – both methods *must* agree numerically because they are identical by construction via (210.11).

Numerical Verification of the Bridge Formula

With $\xi_0 = 4/30000$, $v = 246$ GeV and the Yukawa values from the consolidated table:

Particle	r_i	p_i	$f_i^{\text{(bridge)}}$ from (210.11)	f_i in old table
Electron	$4/3$	$3/2$	$1.485 \cdot 10^7$	$1.468 \cdot 10^7$
Muon	$16/5$	1	$7.146 \cdot 10^4$	$7.099 \cdot 10^4$
Tau	$25/9$	$2/3$	$4.205 \cdot 10^3$	$4.221 \cdot 10^3$

Agreement to $\sim 1\%$ (residual loss from rounding of the input values). Hence (210.11) is numerically confirmed.

Maximal Formula: Connection to the Fine-Structure Constant

The bridge formula (210.11) extends to the **maximal formula** that links masses and the fine-structure constant in one expression (Doc. Mathematische_struktur_De Z. 1019):

$$\boxed{\alpha = c_e \cdot c_\mu \cdot \xi_0^{11/2}} \tag{210.12}$$

with the geometric coefficients (Doc. Mathematische_struktur_De Z. 1034):

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \qquad c_\mu = \frac{9}{4\pi\alpha}, \qquad c_e \cdot c_\mu = \frac{27\sqrt{3}}{8\pi^2\,\alpha^{3/2}} \tag{210.13}$$

Solved for α in closed form (Doc. Mathematische_struktur_De Z. 526):

$$\boxed{\alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi_0^{11/5} \cdot K_{\text{frak}}} \tag{210.14}$$

Numerically: $\alpha^{-1} \approx 137.04$ – consistent with CODATA to 0.013%.

Reading: the fractions $4/3$, $16/5$, $25/9$ in the Yukawa form are *not* simple rational numbers, but disguised expressions in $\alpha, \pi, \sqrt{3}$ (Doc. Mathematische_struktur_De §25.3.1). Reducing the fractions directly destroys this structure. The maximal formula (210.14) shows that the seemingly two independent constants α and ξ_0 are in fact **not independent**, but follow both from the same fractal geometry.

Recommendation for Future Communication

- Use exclusively the Yukawa form $m_i = r_i\,\xi_0^{p_i}\,v$ as the canonical mass formula.
- When the direct form is needed (e.g. for connection to older documents), cite the bridge formula (210.11) explicitly, rather than writing "f_i" without specification.
- Avoid the phrasing "the two methods confirm each other"; say instead "the Yukawa form and the direct form are identical by construction via (210.11)".
- For deeper geometric discussion, cite Doc. Mathematische_struktur_De §25 (maximal formula), where the fractions r_i are decomposed into their α-π-$\sqrt{3}$ structure.

.7 Consolidated Winding Table of the Twelve Fermions

The following table lists, for each of the twelve Standard Model fermions, all four winding layers synchronously. It replaces the scattered and partly conflicting tables in Doc. 005, 006, 042, 202.

Particle	**Spin**(n_θ, n_ϕ)	**Space wind.**(k_x, k_y, k_z)	**Time w.**k_t	n	p_{Yuk}	r_i	$p_{g\text{-}2}$
Charged leptons							
Electron	$(1,1)$	$(1,0,0)$	(mass)	1	$3/2$	$4/3$	base
Muon	$(1,1)$	$(1,1,0)$	(mass)	2	1	$16/5$	$5/3$
Tau	$(1,1)$	$(1,1,1)$	(mass)	3	$2/3$	$25/9$	$4/3$
Neutrinos (double-ξ, $m_\nu = r_\nu \xi^2 m_e$)							
ν_e	$(1,1)$	$(1,0,0)$	(mass)	1	–	1	–
ν_μ	$(1,1)$	$(1,1,0)$	(mass)	2	–	$16/5$	–
ν_τ	$(1,1)$	$(1,1,1)$	(mass)	3	–	$25/9$	–
Up-type quarks ($T_3 = +1/2$)							
Up	$(1,1)$	$(1,0,0)$	(mass)	1	$3/2$	6	–
Charm	$(1,1)$	$(1,1,0)$	(mass)	2	$2/3$	2	–
Top	$(1,1)$	$(1,1,1)$	(mass)	3	$-1/3$	$1/28$	–
Down-type quarks ($T_3 = -1/2$)							
Down	$(1,1)$	$(1,0,0)$	(mass)	1	$3/2$	$25/2$	–
Strange	$(1,1)$	$(1,1,0)$	(mass)	2	1	$26/9$	–
Bottom	$(1,1)$	$(1,1,1)$	(mass)	3	$1/2$	$3/2$	–

Note on the column "Time w.": the temporal winding number k_t is linked via (210.7a) to the mass energy of the mode: $E_t = 2\pi k_t c/L_0$. "(mass)" here means "set by the mass energy"; the corresponding value of k_t is not 0 or 1, but the quantization number of the particle state on the temporal circle direction – in the low-energy unrolling this number appears as the mass m of the particle.

How to read this table:

- Column "Spin": all fermions carry $(1,1)$ – the spin winding only separates bosons from fermions, not generations.
- Columns "Space wind." and n: identical within a generation; they distinguish generations, not particle types within a generation.
- Column "Time w.": carries the mass energy of the mode via (210.7a). For *different* particles in the same generation (e.g. Up vs. Down vs. Charm vs. Strange) the spatial winding is the same, but k_t differs – this is how the temporal circle direction distinguishes the masses.

- Columns p_{Yuk} and r_i: distinguish all twelve fermions from each other; they carry the same information as k_t, in practical low-energy notation.
- Column $p_{g\text{–}2}$: only geometrically well-defined for the charged leptons. For quarks, the anomalous magnetic moment g–2 is not a direct observable (hadronic distribution functions are measured instead); neutrinos carry no electromagnetic $g-2$ contribution. "–" therefore means "not applicable", not "open".

Relation to the Yukawa Form

The Yukawa mass formula $m = r\xi^p v$ from Doc. 006 is the *compact* notation; it is equivalent to the direct geometric form of Doc. 070, where the seemingly simple fractions r are *structured* expressions:

$$\frac{4}{3} = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \qquad \frac{16}{5} = \frac{9}{4\pi\alpha}, \qquad \frac{25}{9} = \frac{27\sqrt{3}}{8\pi\alpha^{3/2}} \tag{210.8}$$

(Doc. 070 §25.3.1 – "numerical fractions must not be reduced directly"). The table above uses the Yukawa form as a practical notation; when communicating to readers expected to be familiar with the underlying geometric structure, reference Doc. 070.

.8 Precision W6: Lepton exponents as a geometric sequence

Affected Documents

- Doc. 116 (Koide formula) §3 "Geometric structure of the exponents".
- Doc. 158 (Koide-to-g-2) §1, §3.2 "$1/3$ step size".
- Doc. 203 (R-operator) §3 "Step size $\Delta p = 1/3$".
- Doc. 202 (Field theory overview) §22.1.

Problem

In several documents the lepton-exponent sequence $(p_e, p_\mu, p_\tau) = (3/2, 1, 2/3)$ is described as an arithmetic sequence with constant step size $\Delta p = 1/3 = 1/D$ (three spatial dimensions). This characterisation is not satisfied arithmetically:

$$p_e - p_\mu \;=\; \tfrac{3}{2} - 1 \;=\; \tfrac{1}{2}, \qquad p_\mu - p_\tau \;=\; 1 - \tfrac{2}{3} \;=\; \tfrac{1}{3}. \tag{1}$$

The steps differ ($1/2$ vs. $1/3$); the claim "$\Delta p = 1/D$" holds literally only for the $p_\mu \to p_\tau$ step.

Doc. 158 (g-2 bridge formula) builds the prediction $\Delta a(\tau\text{–}e) = (144/125)\cdot(m_\tau/m_\mu)\cdot\Delta a(\mu\text{–}e)$ on the assumption of a canonical $1/3$ step size. The bridge formula is numerically correct – but the justification needs to be reformulated.

Binding Reading

The lepton exponents form a **geometric sequence with ratio** $q = 2/3$:

$$p_{n+1} \;=\; \tfrac{2}{3}\,p_n, \qquad \big(p_e,\, p_\mu,\, p_\tau\big) \;=\; \big(\tfrac{3}{2},\, 1,\, \tfrac{2}{3}\big). \tag{2}$$

The ratio $2/3$ is the inverse of the tetrahedral factor $3/2$ of $\xi_0 = (4/3)\cdot 10^{-4}$ (Doc. 180). The non-constant differences $1/2$ and $1/3$ are *consequences* of this geometric sequence:

$$p_n - p_{n+1} \;=\; p_n\cdot(1-q) \;=\; p_n\cdot\tfrac{1}{3}. \tag{3}$$

Specifically: $p_\mu \cdot 1/3 = 1/3$ (for the $\mu \to \tau$ step), $p_e \cdot 1/3 = 1/2$ (for the $e \to \mu$ step).

Connection to Koide $Q = 2/3$

The value $Q = 2/3$ of the Koide relation

$$Q \;=\; \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} \;=\; \frac{2}{3} \tag{4}$$

is *numerically identical* to the sequence ratio $q = 2/3$ from (2). This match is not coincidental: inserting an arithmetic sequence with constant step (e.g. $p = 3/2,\, 1,\, 1/2$ as for the down-type quarks) gives Koide $Q \approx 0.80$; with the geometric sequence $(3/2,\, 1,\, 2/3)$ one obtains $Q = 2/3$. The specific ratio $2/3$ thus appears in two layers of the theory:

- as ratio q of the lepton-exponent sequence,
- as value Q of the Koide symmetry.

Consequences for Doc. 158 (g-2 bridge)

The bridge formula $\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e) = (144/125)\cdot m_\tau/m_\mu$ remains numerically valid. The justification is to be reformulated:

- The $1/3$ difference in the $p_\mu \to p_\tau$ step arises from $p_\mu\cdot(1-2/3) = 1/3$, not from $1/D$.
- The factor $144/125$ comes from the ratio (r_τ/r_μ) of the specific rational prefactors.
- The bridge uses only the $\mu \to \tau$ step (which is why it works without the full sequence structure).

Summary

The characterisation "$\Delta p = 1/3 = 1/D$" used in Doc. 203 §3 and Doc. 158 §1 is to be replaced by the geometric sequence $p_{n+1} = (2/3)\,p_n$. Statements that build on this sequence (Koide Q, g-2 bridge) remain valid. What is corrected is the geometric justification: not "step size $1/D$ from three spatial dimensions", but "geometric ratio $2/3$, inverse of the tetrahedral factor".

Updated Summary

No.	Doc.	Unclear / ambiguous	Binding
W1	Doc. 060, 018, 149	"Winding number of a particle" ambiguous	$f = 1/(4\xi) = 7500$ is the universal base winding density, not particle-specific
W2	Doc. 051, 202	"Winding" without specification	(n_θ, n_ϕ) encodes only spin; all fermions carry $(1,1)$
W3	Doc. 018, 149	Generation hierarchy without winding picture	Generations = fractal branchings with Hausdorff dim. $p_{g\text{-}2} \in \{-, 5/3, 4/3\}$
W4	Doc. 005, 006, 042, 202	Symbol (n,l,j) in three meanings	Wave vector (k_x, k_y, k_z, k_t) on T^4 is primary; (n,l,j) is derived via $n = k_x^2 + k_y^2 + k_z^2$
W5	Doc. 005, 006, 032	Symbol f_i in three meanings; "two methods" presented as independent	Yukawa form $m = r\xi^p\,v$ is canonical; direct form $m = 1/(\xi_0\,f_i)$ identical via bridge formula (210.11)
W6	Doc. 116, 158, 203	Lepton exponents as arithmetic sequence with "$\Delta p = 1/3 = 1/D$"	Geometric sequence $p_{n+1} = (2/3)\,p_n$; ratio $q = 2/3$ identical with Koide $Q = 2/3$

Consolidated winding table for all twelve fermions: see § "Consolidated Winding Table of the Twelve Fermions" above.